Group I

Bright Meal

A brighter plate
for everyone.

THE PROBLEM

1.1M TONNES

Hospitality food waste annually (WRAP, 2024)

2.89M PARCELS

Emergency food parcels distributed (Trussell Trust, 2024/25)

18M TONNES CO2E

From UK food waste each year

THE GAP

Restaurants discard perfectly good food every night. Charities struggle to source it.
No efficient, structured connection exists between surplus and need.

OUR SOLUTION

BRIGHT MEAL

RESTAURANTS POST

Donors list surplus food with allergens, storage, photos, and expiry details

DRIVERS COLLECT

Volunteer drivers claim jobs, pick up donations, and track delivery

CHARITIES RECEIVE

Charities browse offers, accept food, and confirm delivery for impact tracking

Three roles. One platform. Real-time lifecycle from posted to delivered.

FROM PLAN TO PRODUCT

SEMESTER 1

STATIC MOCKUPS
of donor, charity, and driver flows

FIGMA AI
used for initial layout generation

NO BACKEND
no real data, no authentication

Feedback: "No working prototype"

SEMESTER 2

FULL-STACK APP
deployed on Vercel with Supabase

CLAUDE
used as pair-programming assistant

REAL AUTH
RLS security, atomic RPCs, realtime

19 BADGES
gamification, and impact tracking

Three roles. One platform. Real-time lifecycle from posted to delivered.

Every night, restaurants

Bright Meal

TECHNICAL ARCHITECTURE

TECH STACK

FRONTEND
Next.js 16 App Router + Tailwind v4 + shadcn/ui

BACKEND
Supabase PostgreSQL + Auth + Realtime

HOSTING
Vercel (edge deployment, automatic CI/CD)

AUTH
Supabase Auth (email + password, JWT)

STATE
Server Components by default, client for interactivity

SECURITY MODEL

ROW-LEVEL SECURITY
on every table with per-role policies

7 ATOMIC FUNCTIONS
for atomic status transitions

4 ENFORCEMENT TRIGGERS
role lock, status flow, field restrictions, profile lock

SELECT FOR UPDATE
prevents race conditions

NO DELETES ALLOWED
via RLS on any table

Zero raw .update() calls for status changes. Every mutation is an atomic RPC.

GAMIFICATION AND IMPACT

Points and badges

10 POINTS PER KG

(min 5 points) awarded to all three participants on delivery

19 BADGES

across 4 tiers: Bronze, Silver, Gold, Platinum

STREAK BADGES

4 tiers reward weekly consistency

WEEKLY STREAKS

reward consistent participation

REAL IMPACT METRICS

- CO2e savings: 2.5 kg CO2e per kg food rescued (WRAP UK data)
- Meals equivalent: 2.5 meals per kg redistributed
- Live dashboard shows personal and platform-wide impact stats
- Automatic badge awards via SECURITY DEFINER function on each delivery
- Leaderboard with role-based filtering and efficiency rankings

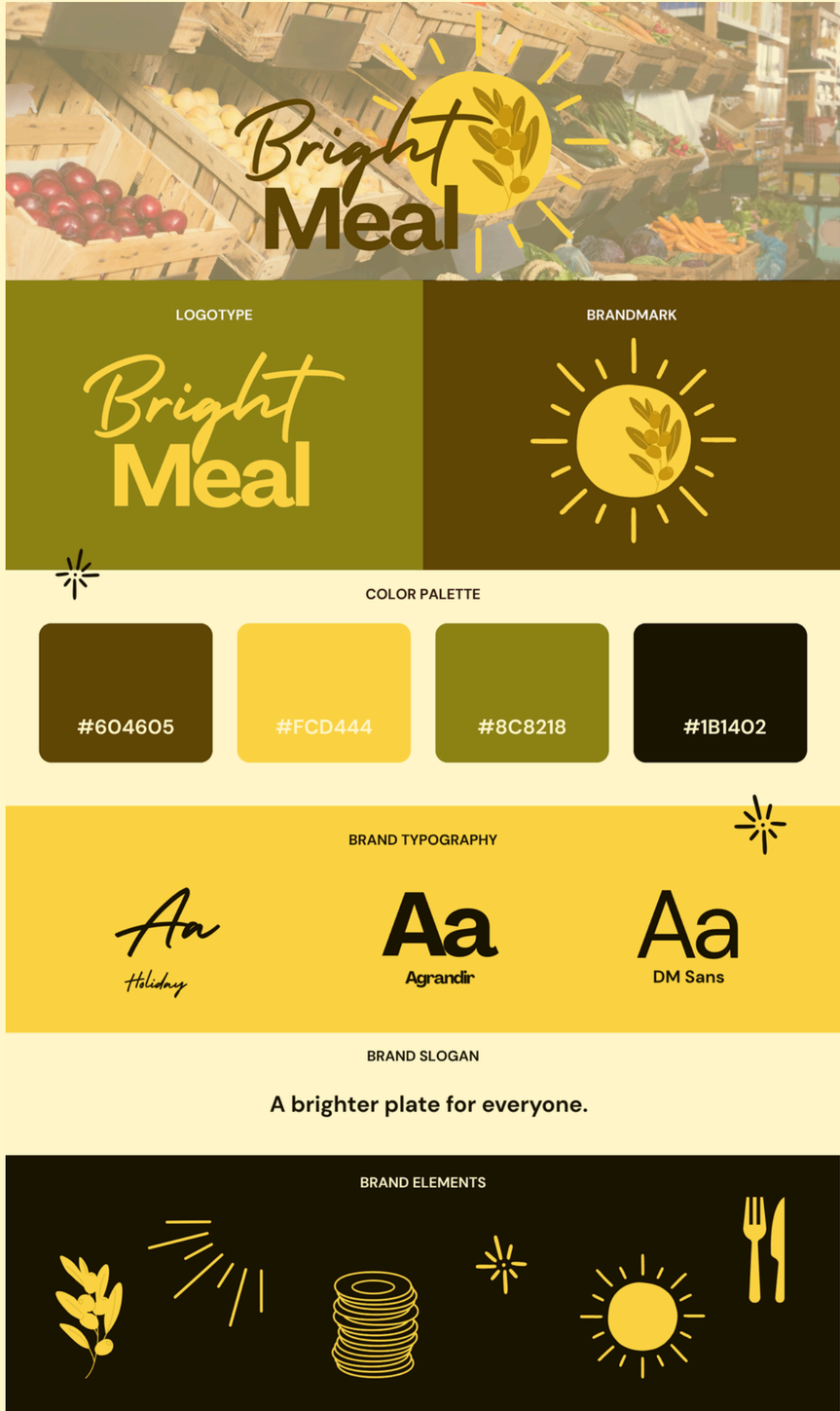
SOME BADGES

- First Steps
- Rising Star
- Legend
- On Fire

BRAND AND OUTREACH

OUTREACH CAMPAIGN

- 16 charities contacted across Bath and surrounding areas
- 6 stakeholder interviews (4 charities + 2 restaurants) and 7 driver walkthroughs across Bath
- Info sheets with app screenshots, impact stats, and testimonials
- Instagram presence for brand awareness and charity engagement
- Direct contacts with food bank managers and community kitchens



TESTING AND EVIDENCE

RESEARCH TIMELINE

SEMESTER 1
Initial surveys, competitor analysis, Figma prototyping

JANUARY - FEBRUARY
16 charities contacted via email and phone

FEBRUARY - MARCH
4 in-depth charity interviews conducted

MARCH
MVP built incorporating all interview feedback

APRIL
2 restaurant interviews +
7 driver walkthroughs +
6 structured usability tests (2 per party) with think-aloud protocol

REAL STAKEHOLDER VOICES

HIVE, BATH
"Public safety has to be the number one priority."

TASTY KITCHEN, BATH
"We throw away a good amount every night - it feels wrong."

COMMUNITY KITCHEN, BATH
"I just take what I get and work out what to cook."

VOLUNTEER DRIVER, BATH
"It feels like Deliveroo but for doing good."

FEEDBACK TO ITERATION

TEST, LEARN, BUILD

COURSEWORK 1 - FEEDBACK

"No working prototype"
Built full MVP on Vercel

"Leaderboard unfair for small donors"
Streak badges + per-role leaderboards added

"Want badges and recognition"
19-badge system, 4 tiers

COURSEWORK 2 - STAKEHOLDER FINDINGS

"Allergen list incomplete -- Natasha's Law requires 14"
Fixed to all 14 allergens

"Trust based on FSA hygiene rating"
Added hygiene rating to profiles

"Want photos to judge food"
Added photo upload field

"Address should auto-fill"
Auto-filled from profile

"Need use-by vs best-before"
Added date type selector

"Want option to collect it ourselves"
Added delivery method toggle (driver/charity)

"Timing section too complex for end-of-day use"
Replaced with 2-tap availability selector + Today button

LEGAL AND COMPLIANCE

REGULATORY FRAMEWORK

FOOD SAFETY ACT 1990

Donors must ensure food is safe. BrightMeal requires allergen, storage, and expiry data on every listing.

UBER V ASLAM [2021] UKSC 5

Drivers are volunteers, not workers. No minimum hours, no obligation to accept jobs. Clear terms of engagement.

UK GDPR / DATA PROTECTION ACT 2018

Supabase RLS ensures data isolation. Users access only their own data. No third-party data sharing.

HEALTH AND SAFETY AT WORK ACT 1974

Duty of care for volunteer drivers. Insurance guidance and safe handling protocols required.

OUR COMPLIANCE ACTIONS

14 ALLERGENS

tracked per Natasha's Law (Allergen Labelling Regs 2021)

FSA HYGIENE RATING

field added to donor profiles for trust verification

ROW-LEVEL SECURITY

on every database table ensures GDPR data isolation

IMMUTABLE AUDIT TRAIL

via donation_events table -- no deletes allowed

VOLUNTEER CLASSIFICATION

documented: no pay, no obligation, opt-in model

ETHICS AND SUSTAINABILITY

ETHICAL PRINCIPLES

Dignity

CHARITIES CHOOSE WHAT FOOD TO ACCEPT. NO FORCED SURPLUS. RESPECTFUL REDISTRIBUTION.

Fairness

EQUAL POINTS TO ALL THREE ROLES PER DELIVERY. SEPARATE LEADERBOARDS AND STREAK BADGES REWARD CONSISTENCY.

Welfare

14 ALLERGENS TRACKED. STORAGE TYPES ENFORCED. SAFE FOOD HANDLING AT EVERY STEP.

CIC STRUCTURE PLANNED:

asset-locked, community benefit, reinvested surplus

SUSTAINABILITY AND SDGS

SDG 12.3: HALVE FOOD WASTE

Direct redistribution from hospitality to charities reduces waste at source

SDG 2: ZERO HUNGER

Connecting surplus food to food banks and community kitchens

SDG 13: CLIMATE ACTION

2.5 kg CO_2e saved per kg food rescued (WRAP UK methodology)

INTERNATIONAL ALIGNMENT

MODEL REPLICABLE GLOBALLY. PLATFORM-AGNOSTIC APPROACH WORKS IN ANY CITY WITH SURPLUS AND NEED.

FORWARD STRATEGY

3-YEAR REVENUE ROADMAP

YEAR 1

Grant-Funded Pilot

CIC registered. Target £20k+ from The Fore, NLCF Awards for All, BCBN. Free for all users. Infrastructure on free tiers.

YEAR 2

Subscription Launch

£25/mo CSR tier for donors: 30 subscribers = £9k revenue. Year 1 grant surplus bridges the £11k operating deficit.

YEAR 3

Self-Sustaining

75 donors (£22.5k) + B&NES council contract at £100/tonne diverted (£12k) = £34.5k total. £850 surplus. Zero grant dependency.

Break-even

Council contract is load-bearing - 75 donors with it vs 113 without. Free forever for charities and drivers.

INTERNATIONAL SCALING

POST-BREXIT EU EXPANSION

UK rules still broadly align with EU, but EU Food Donation Guidelines (2017) impose stricter liability-sharing and traceability. Our structured donation system (use-by, storage, allergens) adapts without redesign.

GDPR TERRITORIAL SCOPE (ARTICLE 3)

Handling EU resident data triggers full EU GDPR compliance regardless of our location — needs EU representative and DPIAs. Data minimisation, encryption, and RLS already lower cross-border friction.

US MARKET: FDA VS FSA

US Bill Emerson Good Samaritan Act shields donors acting in good faith, lowering participation barriers. Our two-stage verification (donor declaration + charity acceptance) works under both regimes.

HONEST REFLECTIONS

01

WE HAD THE HARD SIDE WRONG

We thought convincing restaurants to donate would be the blocker. Charity interviews flipped it. Without allergens, an FSA rating, and a photo, they wouldn't accept at all. The donor form exists to earn charity trust.

02

WE LOCKED FEATURES AND BROKE THE LOCK THREE TIMES

Each time a usability test found something genuinely broken: expired offers, a required field that wasn't required, a button that did nothing. We'd break it again. Scope-locked isn't test-locked.

03

SIXTEEN COLD EMAILS, FOUR REPLIES

Every restaurant we actually walked into said yes. For future cities we'd lead physically, not just through emails.

04

THE COUNCIL CONTRACT IS OUR BIGGEST BET

If it slips to Year 4 we'd need 113 paying donors (17% of Bath) in one year. We'd rather name that fragility than pretend it isn't there.

GENERATIVE AI DISCLOSURE

AI TOOLS USED

CLAUDE (CW2)

Pair-programming assistant for ALL CW2 UI development and code implementation. Used to accelerate Next.js components, Supabase queries, and Tailwind styling.

FIGMA AI (CW1 ONLY)

Used in CW1 only for initial prototype layouts. All CW1 designs were rebuilt from scratch in CW2.

HUMAN-LED WORK

EVERYTHING ELSE WAS US.

All architecture decisions, database schema design, security model (RLS + RPCs + triggers), gamification logic, business strategy, financial projections, charity interviews, outreach messaging, legal analysis, and presentation content were designed and executed by the team. GenAI was not used for business strategy, stakeholder research, or any written analysis.

Bright Meal

A BRIGHTER PLATE FOR EVERYONE.

We built a working platform that connects surplus food to those who need it most.

We're ready to pilot in Bath.